

UnipDroidians 2026 Team Description Paper

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Abstract. Recent advances in domestic robotics have motivated the development of versatile and socially intelligent service robots capable of assisting humans in everyday tasks. This paper presents the UnipDroidians 2026 Team Description Paper (TDP) for the RoboCup@Home Open Platform League (OPL). The team from Universidade Paulista (UNIP), Brazil, developed the UD-H1, a fully customized mobile manipulator designed for domestic services and human-robot interaction (HRI). The system integrates a modular architecture based on ROS 2, combining autonomous navigation, object manipulation, perception, and dialogue management. Key features include a multisensor SLAM approach with LiDAR and RGB-D fusion, a 6-degree-of-freedom (6-DOF) robotic arm with visual servoing, and a bilingual natural-language-processing (NLP) framework supporting communication in Portuguese and English. Experimental evaluations demonstrate reliable navigation and manipulation in semi-structured environments, robust speech understanding under variable acoustic conditions, and fluid multimodal interaction. This research aims to strengthen open-source robotics development within the RoboCup@Home community and contribute reproducible methods for socially aware robotic autonomy. All developed modules are open-source and available at: <https://github.com/roboticaunip>.

1 Introduction

Service robotics has rapidly evolved in the last decade, with increasing emphasis on autonomous systems capable of performing complex tasks in unstructured human environments. The RoboCup@Home initiative offers an international benchmark for evaluating such capabilities, emphasizing real-world applicability, robustness, and human-centered design. Being part of this environment and contributing to its advancement are the main goals of this project.

The UnipDroidians, a novice team, debuted at CBR 2025 (Brazilian Robotics Competition) in RoboCup@Home Brazil with the UD-H1 prototype, an open-platform service robot that enabled learning through iterative development in autonomous navigation, manipulation, and multimodal human-robot interaction. The project leverages the educational and research infrastructure of UNIP's Robotics Laboratory, integrating student innovation with academic supervision to promote applied learning and reproducible science, while also fostering an institutional culture of robotics within a large private university.

For the 2026 edition, the UD-H1 brings improvements to software and hardware subsystems, including a redesigned perception pipeline for greater environmental awareness, a hybrid SLAM framework and navigation using ROS 2 Nav2, and a bilingual dialogue system capable of contextual understanding in Portuguese and English. These advancements contribute to a more adaptive and socially

responsive robotic platform, aligned with the long-term vision of the RoboCup@Home league.

2 Overview

This chapter presents the characteristics of the robot's initial design—a first prototype developed to enable the team's debut in the competition. Despite inherent limitations of the initial version, the project was conceived to serve as a solid foundation for structural, hardware, and software enhancements in future iterations.

The proposed system consists of two manipulator arms mounted on an aluminum-profile chassis, along with a differential-drive locomotion system, forming an autonomous mobile robot oriented toward performing domestic tasks. This first version is composed of a set of fundamental subsystems whose modular organization and integration are illustrated in Figure 1.

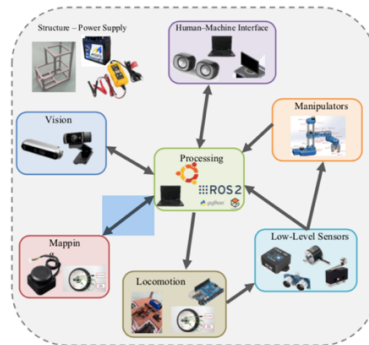


Figure 1 – System Architecture

The developed robot integrates several modules and components, including sensors such as the RPLIDAR C1 (Slamtec), Intel RealSense D457 cameras, IMU, webcam, microphones, and proximity sensors; actuators consisting of 36V brushless hoverboard differential-drive motors, two custom-designed robotic arms with kinematic models developed by the team, and a two-finger gripper with automatic object adaptation; as well as an onboard computing set composed of a Dell laptop, an NVIDIA Jetson board, and an Arduino UNO.

The system runs Ubuntu with ROS 2 Humble and uses frameworks such as ROS, OpenCV, PCL, TensorFlow/PyTorch, and Navigation2. As this is the team's first prototype, there is no comparison with previous versions, but notable features relative to the state of the art include the hoverboard-based motion system and the use of two robotic arms. The mechanics and electronics were designed around this configuration, incorporating 8" brushless motor wheels with solid tires, a dedicated 36V battery, and RioRand ZS-X11H controllers.

The 6-DoF arms follow a kinematic architecture similar to a "1P-5R" arrangement, functionally resembling a SCARA system, equipped with a two-finger compliant gripper. The robot adopts a modular ROS-based architecture with topic-based communication for integration of sensors, motion planning, actuator control, and HRI, as illustrated in Figure 2, which shows the digital model.



Figure 2 – Robot Model in a Digital Environment

2.1 Main Modules and Functions

The system architecture integrates four main modules: Perception and Vision, responsible for image processing and object detection/recognition (Figure 3); Navigation and Localization, including environment mapping, SLAM with Gmapping, global and local planning, and robot pose control; Manipulation, including robotic arm control through ROS, forward and inverse kinematics, grasp planning, and force control using an adaptive gripper; Human–Robot Interaction (HRI), based on speech recognition via `ros_voice_control` and Google Speech API, speech synthesis with Festival, chatbot-style dialogue models, and behavior trees using `ros2_behavior_tree`.

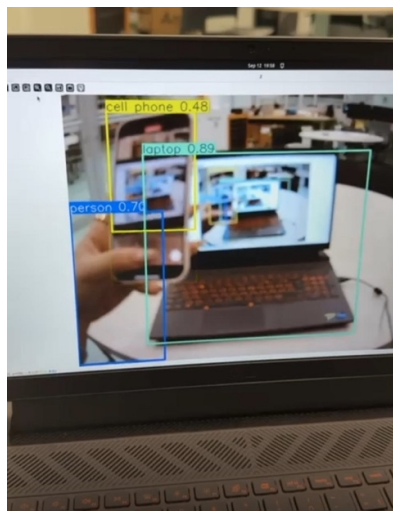


Figure 3 – Object Identification

2.2 Algorithms and Libraries Used

The robot's system is structured into four main modules. In Perception and Computer Vision, the system uses object detection and classification via CNNs such as YOLO, scene segmentation, facial recognition with FaceNet, and sensor fusion between LiDAR, odometry, and camera to improve mapping.

In Navigation and Localization, SLAM is performed using `slam_toolbox` integrating LiDAR and visual odometry, as well as trajectory planning and strategies for avoiding dynamic obstacles using `Navigation2`. As shown in Figure 4, these data feed the Manipulation module, which includes ROS-based arm control using kinematic models, grasp planning, and adaptive-force gripper control.

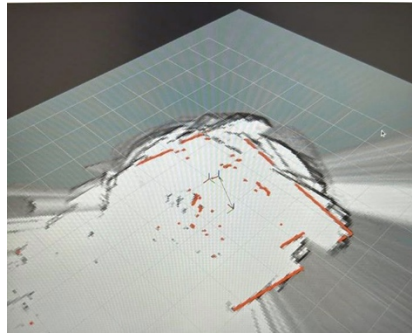


Figure 4 – Mapping Using RPLIDAR and Computing Hardware

Human–Robot Interaction encompasses speech recognition and synthesis, command execution, dialogues, and behaviors using ROS 2 (Figure 5).



Figure 5 – Speech-Based Interaction

2.3 Mechanical and Electrical System

The robot's mechanical structure was built using 30×30 aluminum profiles, arranged in multilayer platforms to accommodate onboard modules, including mounting points for motors, drivers, batteries, odometry sensors, the central computer, communication boards, cable hub, a front object-support platform, and the base for the LiDAR and manipulator arms; an elevated support was also added for the HMI screen and RealSense camera. As this is the first prototype, the main goal was to develop a modular robotic platform that allows continuous evolution and future expansion of capabilities, as shown in Figure 6.

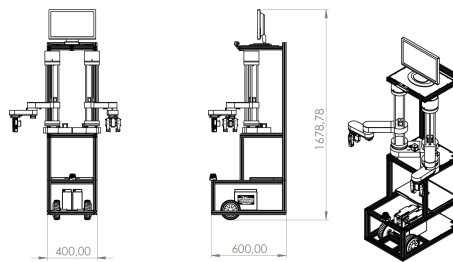


Figure 6 – Views of the Robot

3 System Architecture and Evolution of the UD-H1

The UD-H1 platform was designed around a modular and distributed architecture built on ROS 2 (Humble Hawksbill), aiming for scalability, reuse, and fault tolerance. The system integrates multiple subsystems—perception, navigation, manipulation, and interaction—under a unified task-coordination layer based on behavior trees.

After the first version presented in the national competition, the UD-H1 underwent structural and software improvements that refined its architecture. Enhancements include reorganizing distributed nodes, optimizing perception and navigation pipelines, tuning manipulation controllers, and advancing the HRI layer. These updates strengthened system robustness, increased operational consistency, and enabled tighter integration between modules, preparing the platform for continuous evolution.

3.1 Navigation and Localization

The UD-H1 employs a hybrid localization approach, combining LiDAR-based SLAM with visual SLAM (ORB-SLAM3) for greater robustness under variable lighting and partial occlusions. Fusion of 2D laser and RGB-D camera data is performed using an Extended Kalman Filter (EKF). The navigation stack was built on the Nav2 framework, incorporating a custom local planner optimized for narrow passages and dynamic environments with pedestrian traffic. Improvements after the initial version included tuning planner sampling parameters and refining maps, resulting in smoother and more predictable navigation.

3.2 Manipulation

The modified SCARA-type 6-DoF arm is controlled using MoveIt 2, enabling advanced motion planning, collision avoidance, and real-time trajectory adaptation. A visual-servoing module corrects the end-effector position during grasping to compensate for sensor drift and kinematic uncertainties. Object grasping is guided by a perception pipeline based on YOLOv8, integrated with geometric algorithms for 3D pose estimation. This increases accuracy in pick-and-place tasks involving objects of varying dimensions and materials. Post-competition improvements included optimized inference times and recalibration parameters, enhancing the stability of manipulation tasks.

3.3 Human–Robot Interaction (HRI)

Interaction between the UD-H1 and users occurs through a multimodal interface combining speech, gestures, and dialogue context. Speech recognition uses Vosk ASR with bilingual (Portuguese/English) models trained for moderate-noise environments. Intent interpretation uses Rasa NLU paired with a custom context manager able to maintain conversation state across multiple commands. Gesture recognition with OpenPose enables detection of simple hand movements and body posture, expanding non-verbal communication. Post-competition improvements included threshold tuning and refinement of intent-handling rules to reduce ambiguities.

3.4 Software Infrastructure

Each subsystem operates in isolated Docker containers, supporting reproducibility, independent maintenance, and rapid deployment. Communication between modules uses ROS 2 DDS for real-time messaging. The overall architecture was strongly designed with educational goals in mind: enabling each subsystem to be studied, modified, and extended by students and researchers, while maintaining competitiveness under RoboCup@Home constraints.

4 Experiments and Results

Experiments aimed to evaluate autonomy, perception, and interaction capabilities of the UD-H1 in conditions similar to typical RoboCup@Home domestic environments. Tests were conducted both in simulation—using Gazebo 11 integrated with ROS 2—and on the physical platform in an indoor space configured with home-inspired layouts. After the national competition—where hardware and integration limitations prevented full task execution—the team implemented improvements described in the previous chapter and conducted a new series of experiments. Main results follow below.

4.1 Navigation and Mapping

In 25 trials involving static and moving obstacles, the UD-H1 completed 82% of planned routes. The mean positional error estimated by the EKF ranged from 8 to 12 cm, suitable for indoor navigation. The system safely navigated corridors as narrow as

70 cm and avoided collisions with simulated pedestrians in 76% of attempts. These results reflect gains from planner optimization and map refinement.

4.2 Object Recognition and Manipulation

The computer-vision system recognized 25 classes of household objects with an average accuracy of 81%. In manipulation tasks, grasp-success rate was 54%, varying with object shape, size, and texture. The visual-servo routine reduced end-effector error to approximately 3.2 cm, contributing to more stable pick-and-place execution. These values represent improvements compared to performance at the national stage.

4.3 Speech and HRI Evaluation

In low-noise environments (< 40 dB), the ASR module achieved 84% accuracy in recognizing voice commands. Under moderate noise (~ 60 dB), accuracy dropped to 68%, consistent with offline ASR models. Rasa NLU correctly interpreted 72% of compound and sequential commands. RGB-D-based gesture recognition achieved 65% reliability, impacted mainly by lighting variability and partial occlusions.

4.4 Overall System Performance

In a set of 30 multi-stage household tasks combining navigation, perception, and manipulation, the UD-H1 achieved an overall completion rate of 41%. Performance varied according to scene complexity, number of objects, and time constraints. Despite remaining limitations, results show that the UD-H1's modular and distributed architecture provides a solid foundation for incremental capability growth, allowing continuous improvements to be safely and systematically integrated.

5 Conclusions

The experiments demonstrate significant evolution of the UD-H1 compared to the prototype presented in the national stage, with structural, software, and integration improvements described throughout this chapter. Although limitations remain in multi-stage tasks requiring fine coordination between navigation, perception, and manipulation, the system showed consistent performance in controlled scenarios and proved capable of executing fundamental RoboCup@Home subtasks. Navigation became more stable, manipulation more precise, and HRI more responsive, solidifying the distributed modular architecture as a strong foundation for future iterations.

The 2026 edition of the UD-H1 consolidates this progress, representing the UnipDroidians' continuous effort to build an open, educational, and competitive domestic-service robot. The system integrates key aspects of modern robotics research—multimodal perception, bilingual natural-language interaction, hybrid navigation, and adaptive manipulation—into a coherent structure suited for real domestic environments.

Experimental validation reinforced the platform's ability to perform autonomous navigation, object manipulation, and natural human–robot interaction within RoboCup@Home constraints. These results demonstrate not only increasing

maturity of the software architecture and sensor pipelines, but also the educational impact of involving students at multiple levels in developing a complete robotic system—from hardware to final integration in real environments.

Based on these advances, future research will follow three main axes: Incorporating reinforcement learning to improve task sequencing and dynamic adaptation; Implementing semantic mapping to enable high-level spatial reasoning and persistent memory; Developing cloud-collaborative learning, allowing multiple robots to share knowledge through federated databases within UNIP's robotics infrastructure.

By maintaining active participation in RoboCup@Home, the UnipDroidians aim to strengthen Brazil's presence in global service robotics and promote open-source, collaborative practices within the international research community. The UD-H1, in its continuous evolutionary cycle, represents not only a robot but a growing scientific and educational ecosystem.

6 Acknowledgments

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Robot UD-H1 [OPL]

The UD-H1 robot at CBR 2025 (Brazilian Robotics Competition) of RoboCup@Home – Brazil.



Table with the characteristics of the prototype in its first version.

Robot version	2025
Dimension LxPxA	400mm x 600mm x 1679mm
Total weight	48kg
Driving motors	brushless 36V
Control Board	RioRand ZS-X11H
Wheel diameter	8 pol
Max Speed	< 1m/s - (limited by the board)
Battery 1	Li-ion 36V
Battery 2	VRLA 12MVA18 - 18Ah
Autonomy	~ 1 hour

Hardware Description

Specifications are as follows:

Battery Types

12V Stationary Battery: Supplies power to the logic system and robotic arms through voltage regulators.

36V Lithium-ion Battery: Provides power to the drive wheels.

Base Actuators

Two 6.5" Hoverboard Motors (with Hall sensors).

Each motor is controlled by a RioRand ZS-X11H driver board.

Fully integrated with ROS 2 for navigation control.

Torso Actuators

Not applicable (no torso actuators implemented).

Head Actuators

Fixed head, no motion.

(Future work: addition of a pan-tilt mechanism.)

Arm Actuators

SCARA-type robotic arm powered by NEMA 17 stepper motors.

The gripper is actuated by a servo motor.

Currently using one SCARA arm.

(Future work: two SCARA arms for full bimanual operation.)

Other Actuators

Not applicable.

Laser Sensors

2× RPLIDAR C1 used for SLAM and mapping (ROS 2 Cartographer integration).

RGB Sensors

USB Webcam for standard image and video capture.

RGB-D Sensor

Intel RealSense D435 for 3D perception, obstacle avoidance, and navigation.

Other Sensors

Micro switches acting as bump sensors.

HC-SR04 ultrasonic sonar.

Sharp GP2D120 infrared sensor for short-range distance.

(Future: 9-DOF IMU for improved odometry fusion.)

Display

15" Touchscreen Panel.

Additional Dell laptop display used for extended processing and control.

Speakers

HP Speaker Box for audio output and interaction cues.

Embedded System

NVIDIA Jetson running Ubuntu 22.04 + ROS 2 Humble

Executes SLAM, Nav2, and AI/vision modules.

Arduino Uno + CNC Shield for SCARA arm control.

Arduino Mega (optional) for additional sensory modules.

Also our robot incorporates the following devices:

Auto-focus all-purpose cameras.

Battery charge indicator.

Software Description

For our robot we are using the following software:

Operating System

Ubuntu 22.04 LTS with ROS 2 Humble.

Middleware

ROS 2 communication framework.

Key packages: cartographer_ros, nav2, realsense2_camera (future), yolov8_ros (future)

Navigation

SLAM with Cartographer using RPLIDAR C1.

Current implementation still requires optimization for improved map quality.

Nav2 available for global/local planning, but full integration is planned for later stages.

Localization

AMCL fused with LIDAR data for robust localization. *(Future)*

Sensor fusion using IMU + RealSense depth to handle drift and occlusions. *(Future)*

Mapping

Cartographer currently provides the most stable 2D mapping results.

(Future) Integration of RealSense D435 depth data for 2.5D mapping, allowing detection of tables, shelves, and elevated obstacles.

Object Recognition

YOLOv8 running on the Jetson and integrated with ROS 2.

Supports real-time detection of people and common household objects.

Manipulation

One SCARA arm with servo-based gripper.

(Future) Two SCARA arms fully integrated into ROS 2 MoveIt or custom trajectory planning.

Human Detection

Hybrid approach:

LIDAR for close-range proximity detection

RGB + YOLO for human detection under varying light conditions.

Face Detection

OpenCV HOG-based face detector using the main RGB camera. *(Future)*

Face Recognition

FaceNet or Python face_recognition library for person identification. *(Future)*

Gesture Recognition

OpenCV + MediaPipe for hand and body gesture tracking integrated into ROS 2.

Speech Synthesis

Festival or eSpeak for local TTS. *(Future)*

Azure TTS for higher-quality cloud-based speech. *(Future)*

Speech Recognition

Google STT for online mode; PocketSphinx as offline fallback. *(Future)*

Natural Language Processing (NLP)

SpaCy or Rasa for command interpretation and dialog management. *(Future)*

External Devices

UD-H1 robot relies on the following external hardware:

USB Joysticks for teleoperation and debugging.

Cloud Services

UD-H1 connects the following cloud services:

Microsoft Azure for offloading demanding AI tasks (advanced NLP, large-scale vision).

Local processing remains primary to ensure offline autonomy.